

Week 10 - 4 December

Section outline

- Date: **Thursday, 4 December**

Time: **14:00 - 18:00**

Location: **PR_B501D**

Lecturer: Zhou Zhou

Session Content

- Transfer learning concepts
- Using pre-trained CNNs in PyTorch (Colab)
- Creative robotics examples
- Assignment 4 (Detection System)
- Mini-project discussion & support
- Slides: [Week10_Transfor_Learning_Slides](#)

Quick explanation for today's content: ["Fine-tuning a Neural Network explained" by deeplizard](#)

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Colab 1: Transfer Learning BasicsURL

In this notebook you will:

1. Load a **pre-trained ResNet18** model in PyTorch
2. Freeze its layers and replace the final classification layer
3. Train the new classifier on the CIFAR-10 dataset
4. Think about how this kind of model could be used in a creative robotics context

💡 Don't worry if some of the code looks long – you mostly just need to **run cells** and then **experiment with a few marked parameters**.

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Colab 2: Feature Extractor vs Fine-TuningURL

In this notebook you will:

1. Train **Experiment A** – ResNet18 as a frozen feature extractor
2. Train **Experiment B** – ResNet18 with last block + classifier fine-tuned
3. Compare training and test performance
4. Reflect on which strategy might be better for a **robotic system**

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Colab 3: Detection Pipeline Starter (Assignment 4)URL

In this notebook you will:

1. Install and load a **pre-trained YOLO** object detector
2. Run detection on a sample image
3. Extract detected labels from the model output
4. Design a **simple "robot brain"** that reacts to detections

This is a **starting point for Assignment 4**, where you'll build your own detection-based system.

Assignment 4:

Goal: Implement pre-trained models for object/face detection

Task:

5. Create a detection system using YOLO or face detection.
6. Integrate with robotics application concept.

- Pre-trained models you might wanna use:

ResNet (Residual Networks): The standard workhorse. Good balance of accuracy and speed.

VGG16: Older, simpler, but very heavy (large file size).

MobileNet (Crucial for Robotics):

Optimized for mobile devices and embedded systems (Raspberry Pi/Jetson Nano).

Slightly less accurate, but much faster inference (FPS).

YOLO (You Only Look Once): The gold standard for Object Detection (finding where things are).

